

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Winter Examination – 2024 Course: B. Tech Branch: Petrochemical Engineering Semester: VI Subject Code & Name: BTHM604 Process Economics and Industrial Management Max Marks: 60 Date: 26/12/2024 Duration: 3 Hr.				
Instructions to the Students: 1) Each question carries 12 marks. 2) Question No. 1 will be compulsory and include objective-type questions. 3) Candidates are required to attempt any four questions from Question No. 2 to Question No. 6. 4) The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 5) Use of non-programmable scientific calculators is allowed. 6) Assume suitable data wherever necessary and mention it clearly.				
			(Level/CO)	Marks
Q.1	Objective type questions. (Compulsory Question)			12
1	The cost independent of on the volume of production and sales of a product is called its _____.			R/CO1
	a. Fixed cost	b. Variable cost	c. Total cost d. None of these	
2	Which of the following does not come under the sales expenses for a product of a chemical plant?			U/CO1
	a. Advertising	b. Warehousing	c. Legal fees d. Customer service	
3	The flow back of profit plus depreciation from a given project is called_____			R/CO1
	a. capital recovery	b. cash flow	c. economic return d. earning value	
4	The cost varies as per the time and regular need known as_____.			U/CO1
	a. Operating Cost	b. Overhead cost	c. Direct production cost d. General expenses	
5	The difference between the present and future worth of money at some time in the future is called_____			A/CO1
	a. Discount	b. Deduction	c. Inflation d. Depletion	
6	The interest calculated on the basis of 365 days a year, is known as_____.			A/CO1
	a. Interest	b. Ordinary simple interest	c. Exact Simple Interest d. None of these	
7	What is the type of annuity that does not have a fixed time span but continues indefinitely or forever?			R/CO1
	a. ordinary annuity	b. perpetuity	c. Annuity due d. deferred annuity	
8	Break-even point increases with_____			A/CO4
	a. Increase in fixed cost	b. Increase in variable cost for a given value of price	c. Decrease in unit contribution d. All of these	
9	Project Scheduling provides information about the_____			U/CO5
	a. Proper utilization of machines	b. Means to minimize idle time for machines	c. Time of completion of job d. Time of starting of job and also about how much work should be completed during a particular period	

10	Gantt chart (or Bar chart) is helpful in _____				A/CO5	1
	a. Efficient utilization of manpower and machines	b. Preparing production schedule	c. Efficient dispatching of products	d. Inventory control		
11	In project management when does planning take place _____				U/CO5	1
	a. before the project	b. during the project execution	c. at the start of the project	d. after the project		
12	PERT stands for _____.				R/CO5	1
	a. Program Estimation and Reporting Technique	b. Process Estimation and Review Technique	c. Program Evaluation and Review Technique	d. Planning Estimation and Resulting Technique		
Q 2	Solve the following.					12
A)	The purchase cost of 50gal glass-line jacketed reactor was Rs.6, 85, 906.58 in 1981.Estimate the purchased cost of a similar 300gal glass-line jacketed reactor (Rs.) in 1986 using Annual Average Marshall & Swift Equipment Cost Index (All Industry) to update a purchase cost of reactor. Where, Cost Index for year 1981 & 1986 are 721 & 798 respectively.				U, A/CO1	6
B)	Explain the Cash Flow for Industrial Operation with neat sketch.				R, U/CO1	6
Q. 3	Solve the following.					12
A)	A heat exchanger is to be used in heating process. A standard type of heat exchanger with a negligible scrap value cost of Rs. 16, 42,890 and will have useful life of 6 years. Another type of heat exchanger with equivalent design capacity is priced at Rs. 27, 92,913 but with a useful life 10years and a scrap value of Rs. 3, 28,578 Assume an effective compound interest rate of 6% per year Determine which heat exchanger is cheaper by comparing the capitalized cost of each.				U, A/CO2	6
B)	Differentiate between PERT and CPM analysis (At least 6 point)				R, U/CO5	6
Q.4	Solve Any Two of the following.					12
A)	A plant requires initial fixed capital of 9Lakh and 1Lakh of working capital it is essential that the annual income will be 8 lakh and the annual expenses including depreciation of 5.2Lakh before tax. Income tax amount to be 34% of all free taxes profit. Calculate i) Annual Rate of Return on Total Capital Investment Before Income tax ii) Annual Rate of Returns on Fixed Capital after Income Tax.				A/CO4	6
B)	The cost of 2-independent process variables F1 & F2 affects Total Cost (C _T) in Rs. of the process as per the following equation. $C_T = 100F_1 + 1000/F_1 F_2 + 20F_2^2 + 5$ Find the Total Cost Optimum value C _T				R/CO4	6
C)	Derive the book value equation using Sinking Fund method. $V_a = V - (V - V_s) * \{((1+i)^a - 1)/((1+i)^n - 1)\}$				R,U/CO3	6

Q.5	Solve Any Two of the following.		12
A)	Explain in detail about “Life Cycle of Project”.	R,U/ CO5	6
B)	Capitalized cost is defined as the sum of the original cost (C_v) of equipment plus the amount (P) that must be invested when the original equipment is purchased so that when the original equipment or asset is replaced in (N) years at a cost (C_R), the value of the investment equal (P) plus (C_R). Derive two expressions for capitalized cost based on (i) Annual discrete interest compounding and (ii) Continuous interest compounding.	R, U/ CO2	6
C)	Heat integration unit is planted in plant at investment of 25 Lakh this took result into energy saving of 20GJ/yr. The interest rate is 15% per annum. Plant life is 3 years, find break even cost of energy in Rs/GJ ? Assume the initial investment for system arrangement equal to saving total cost.	U, A/ CO4	6
Q.6	Solve Any Two of the following.		12
A)	A bond matures after 5 year and has maturity value of Rs.1000. If interest rate is 12 % . What is the Present Worth of bond and calculate the Discount? And Explain Annuity term.	R, U/ CO2	6
B)	Initial cost of chemical plant is 72 lakh & salvage value of 2 Lakh, with having service life of 7 years. Calculate book value & depreciation value at the end of 4 years using sum of the year digit method.	U, A/ CO3	6
C)	The Original value of equipment Rs.22000/- and scrap value is Rs.2000/- at the ends of its useful life period of 10 years. Calculate an asset value using Straight Line at the end of each year and plot the graph of Asset value vs Life period using graph paper.	U,A/ CO3	6
	*** End ***		